The Song Before Civilisation

Das Lied vor der Zivilisation

Energetic, Cognitive, and Medium Constraints on Marine Megafaunal Intelligence, and the Archaeocete Observer Paradox

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Abstract

We present a theoretical–ecological framework for a long–misstated question: why do several marine mammalian lineages of demonstrably high cognitive capacity not converge on cumulative, artifact–based technological civilisation? Our central claim is that the apparent paradox is largely an *artefact of the observer's coordinate frame*. Terrestrial cognition silently substitutes "history" with "externalisable, artifact–based record," and thereby misreads the non–terrestrial *media* of oceanic history as an *absence* of history. We formalise the system through three coupled state variables—metabolic scaling M, cognitive complexity I, and environmental information persistence E—and give three formal upgrades: (i) a macroevolutionary ODE flow (M, I, E, C) now driven by *glacial resource–collapse forcing*; (ii) an agent–based model of whale, shark and primate lineages, now with *horizontal, between–clan dialect transmission*; and (iii) an order–parameter theory in which civilisation is a topological phase transition in M–I–E space. All three converge on one result: the ocean does not abolish history—it stores it primarily in *embodied–acoustic* form (dialect, matriline, song, migratory route, prey tradition), and because durable external substrate $S_{ext} \to 0$, cumulative artifact culture C_a cannot ignite. We ground the framework in the empirical literature on cetacean culture and close with falsifiable predictions.

Keywords: theoretical ecology; macroevolutionary dynamics; cetacean culture; cultural transmission; phase transition; observer bias; embodied–acoustic history

1 Introduction: a paradox manufactured by the observer

The classical statement runs: odontocetes and archaeocetes evolved high intelligence, high sociality, and elaborate communication, yet produced no human–style technological civilisation—hence a paradox. This framing smuggles in a very strong premise,

$$\text{Intelligence} \implies \text{artifact–based civilisation}, \tag{1}$$

i.e. that high intelligence *ought* to externalise into tools, fire, symbol, metal, architecture, and archaeologically recoverable remains. That criterion is drawn entirely from terrestrial primate experience. When oceanic intelligence fails to follow this path, the terrestrial observer reads "absence," "failure," "stasis." This is to *judge the geometry of the deep sea with a ruler cut for the trees*.

We formalise this misreading as an observer–operator bias. Let I_M be the true structure of the marine intelligence system and O_T the terrestrial observer's interpretive operator; the "paradox" produced is

$$P \;=\; O_T(I_M) - I_M. \tag{2}$$

The point of (2) is that P need not originate inside I_M at all; it originates in O_T. When the observer treats "external artifact accumulation" as the default terminus of intelligence, any non–artifactual intelligence is projected as paradox (Fig. 4). The empirical record supports this reading: cetacean culture is not a metaphor but an active object of behavioural–ecology research [1, 2].

2 History decomposed

The deepest flaw of the original framing is the identification of "history" with "human archive." If history is defined as *persistent cross–generational state–dependence*, the ocean plainly has history. We decompose it:

$$H \;=\; H_p + H_c + H_e + H_a, \tag{3}$$

with components as in Table 1.

Symbol	Meaning	Oceanic carrier	Terrestrial carrier
H_p	phylogenetic history	*Basilosaurus*→*Physeter*	primate lineage
H_c	cultural history	dialect, song, prey tradition	language, ritual
H_e	ecological history	migratory route, prey specialisation	settlement, agriculture
H_a	externally inscribed history	*relatively weak*	tool, fire, symbol, architecture

Table 1: The four–fold decomposition of history. Key correction: cetaceans do not lack H; they have weak H_a but strong H_p, H_c, H_e.

Sperm whales furnish the clearest case: their largest affiliative structures, the *vocal clans*, are marked on ocean–basin scales by culturally transmitted coda dialects, with membership and behaviour maintained largely through social learning within matrilineal structure [3–5]. The difference between oceanic and terrestrial history is therefore not "history vs. no history" but

$$\underbrace{H_{\text{sound}} + H_{\text{route}} + H_{\text{kin}} + H_{\text{prey}} + H_{\text{body}}}_{\text{embodied–acoustic history } H_M} \quad \text{vs.} \quad \underbrace{H_{\text{artifact}} + H_{\text{fire}} + H_{\text{symbol}} + H_{\text{arch}}}_{\text{artifact–dominant history } H_T}.$$

The valid inference chain is weak artifact persistence ⇒ limited artifact–based cumulative technology, and *never* ⇒ no history.

3 System definition and scaling laws

Define the state vector $S = (M, I, E)$ and cumulative culture C.

Metabolic–cognition coupling. $I \propto M^{\alpha}$, $\alpha \in (0.2, 0.5)$: body size raises the neural ceiling with diminishing returns.

Energetic instability. $E_{\text{var}} \propto M^{\beta}$, $\beta > 1$: gigantism amplifies energetic volatility, rendering large lineages acutely sensitive at resource troughs.

Cultural–memory constraint. $C(t) = \int_0^t f(I, S_{ext})\, dt$, with marine constraint $S_{ext} \approx 0$, so $dC/dt \to 0$ in the long run. Equivalently, splitting total culture into embodied and artifact parts, $C = C_b + C_a$ with cetaceans $C_b \gg C_a$ and humans C_a undergoing runaway accumulation.

4 Upgrade I: macroevolutionary ODE with glacial forcing

We extend the state to (M, I, E, C) and, following the reviewer's request, add a Milankovitch–like *glacial resource–collapse forcing*. A resource multiplier

$$R(t) = 1 + A\left[0.5 \sin \tfrac{2\pi t}{T_1} + 0.3 \sin \tfrac{2\pi t}{T_2} + 0.2 \sin \tfrac{2\pi t}{T_3}\right], \quad (T_1, T_2, T_3) = (100, 41, 23)\,\text{kyr}, \tag{4}$$

modulates both carrying capacity and energetic variance:

$$\dot{M} = r_M M\left(1 - \frac{M}{K_M R_c}\right) - c_{ME}\, E_{\text{var}}\, M, \tag{5}$$

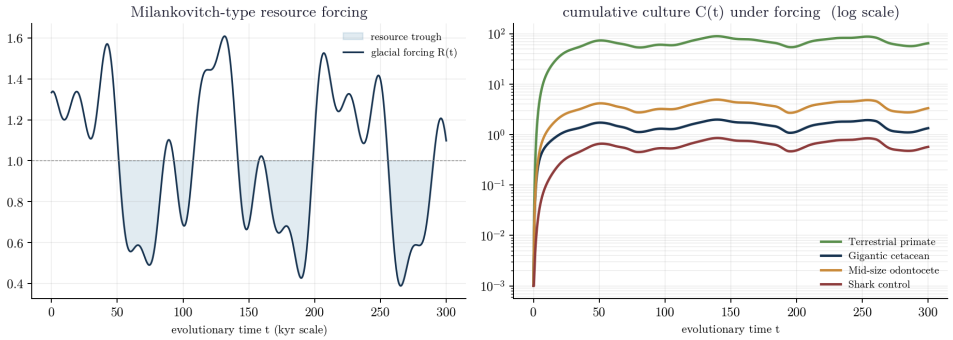
$$\dot{I} = a\, M^{\alpha} - g\, E_{\text{var}} - d_I\, I, \qquad E_{\text{var}} = \varepsilon_0 M^{\beta}/R_c, \tag{6}$$

$$\dot{E} = -k_E E + s_{ext}, \tag{7}$$

$$\dot{C} = f_{\max}\, \frac{I^2}{I_h^2 + I^2}\, E\, R_c - \mu C - \kappa\, \max(0.6 - R_c, 0)\, C, \tag{8}$$

where $R_c = \max(R, 0.15)$ is a resource floor and the last term in $\dot{C}$ is the *trough–reset*: at glacial minima, embodied culture is culled. Terrestrial lineages carry a small κ (artifact buffering); gigantic marine lineages carry large β and weak buffering. Numerically (Fig. 1), glaciation depresses culture in every regime but proportionally hardest in marine giants, whose C is repeatedly reset at each resource minimum, whereas the terrestrial line largely rides through troughs on its artifact reservoir.

Fig. 5 — Glacial resource-collapse forcing: gigantic marine culture repeatedly reset at troughs



Terrestrial lineages ride through troughs on their artifact buffer; gigantic cetaceans (high β, weak buffering) have C cut down at every resource minimum. The log axis makes the terrestrial–marine gap and the trough resets both visible.

Figure 1: Glacial resource–collapse forcing. Left: the Milankovitch–type multiplier $R(t)$ and its troughs. Right: cumulative culture $C(t)$ under forcing—the terrestrial reservoir persists through minima, the gigantic marine trajectory is repeatedly cut down.

5 Upgrade II: agent–based model with horizontal dialect transmission

The ABM now represents each species as a set of *clans*, each carrying a dialect vector $\mathbf{d}$ on a ring ocean, plus member individuals with energy and embodied memory. Two transmission channels operate. *Vertical*: offspring inherit their natal clan's dialect and learn behaviour within the clan. *Horizontal*: spatially overlapping (sympatric) clans pull their *non–identity* dialect dimensions toward one another at rate h, while *identity* dimensions resist as symbolic markers. This is directly grounded in the data: sperm–whale non–identity codas grow more similar between sympatric clans, indicating social learning across symbolic cultural boundaries [5, 6]; humpback song "revolutions" propagate horizontally west–to–east across the entire South Pacific basin [7–9]; and resident killer–whale dialects change partly by inter–matriline horizontal matching within a nested matrilineal society whose pod repertoires remain stable for decades [10–12].

Crucially, dialect remains an *embodied* substrate: when a clan goes extinct, its dialect is erased—there is no artifact record—so horizontal transmission redistributes culture without ever converting C_b into cumulative C_a. Figure 2 shows the result: with horizontal transmission off, between–clan dialect diversity stays high; switching it on drives non–identity dimensions to convergence while identity dimensions persist as clan markers, exactly the empirical signature.

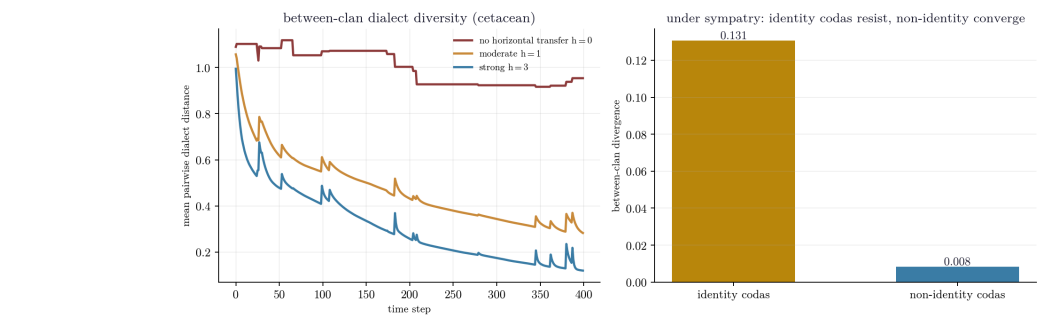


Figure 2: Horizontal (between–clan) dialect transmission. Left: between–clan dialect diversity collapses as horizontal rate h increases. Right: under sympatry, identity codas resist convergence (symbolic markers) while non–identity codas converge—cf. [5, 8].

An earlier agent–based model of killer–whale dialects established that a null model *without* vocal learning cannot reproduce observed group–specific repertoires [13]; our model reproduces both the maintenance of clan identity and the sympatric convergence of shared material, and embeds them in the energetic–medium constraint that forbids artifact accumulation.

6 Upgrade III: civilisation as a topological phase transition

On the (I, E) plane, take the steady–state cumulative culture C^* as order parameter. Setting $\dot{C} = 0$ gives the linear steady state $C^* = f_{\max} f(I) E/\mu$; when external medium is self–catalytic (more artifact $\Rightarrow$ easier preservation), the self–consistent equation

$$C = \frac{k f(I)\, E}{1 - k f(I)\, \rho}, \qquad k = f_{\max}/\mu, \tag{9}$$

jumps where the denominator crosses zero—this is the civilisation transition. The order–parameter map (Fig. 3) exhibits three attractors: the shark type (low I, low E), the cetacean high–intelligence type (high I, low E; confined to the embodied–acoustic phase, C_a bounded), and the terrestrial civilisation type (high I, high E; $C \to$ runaway). The marine constraint band $E \approx 0$ imprisons oceanic lineages in the dark phase no matter how high I climbs.

7 Main theorem

Theorem 1 (Cetacean Non–Cumulativity)**.** *In marine megafaunal systems, high cognitive capacity does not imply cumulative technological evolution. If* (i) $S_{ext} \to 0$*;* (ii) E_{var} *increases with* M*; and* (iii) *the growth of* C *requires stable external encoding, then* $\lim_{t\to\infty} C(t)$ *is bounded: intelligence saturates, culture is periodically reset, and no transition to technological civilisation occurs.*

Proof sketch. From $\dot{C} = f_{\max} f(I) E - \mu C$ and the steady state $E^* = s_{ext}/k_E \to 0$, we obtain $C^* = f_{\max} f(I) E^*/\mu \to 0$ as a bounded supremum. Because $E^* \to 0$, the self–catalytic term cannot drive the denominator through zero, so the runaway branch is unreachable. □ □

Proposition 1 (Terrestrial Observer Bias)**.** *The apparent absence of cetacean civilisation is partly an artefact of the observer operator:* O_T *systematically amplifies the artifact dimension* H_a *and suppresses* H_c, H_e*, making* $P = O_T(I_M) - I_M$ *significantly positive (Fig. 4).*

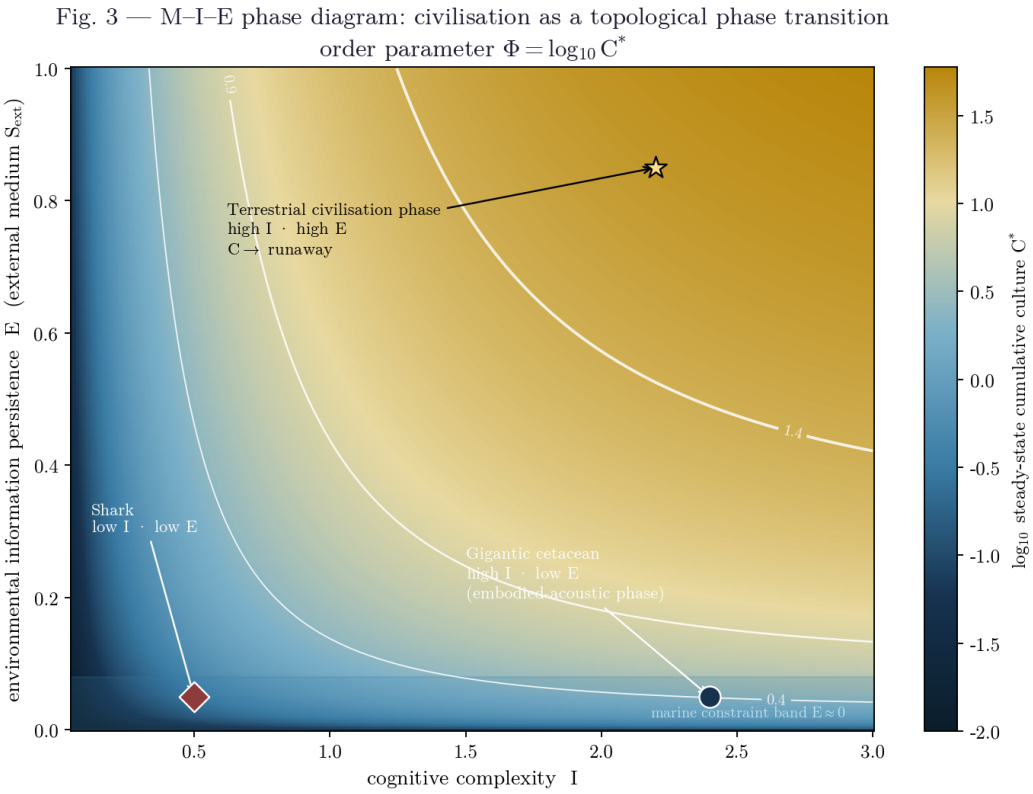


Figure 3: The *M–I–E* phase diagram. Civilisation is a topological phase transition in state space; the marine band $E \approx 0$ (bottom) confines oceanic lineages to the embodied–acoustic phase.

8 德文小结 — Deutsche Zusammenfassung

Der Kern des Arguments lässt sich knapp fassen. Das Meer unterdrückt nicht die Geschichte; es verlagert lediglich ihr *Medium*. Wale bewahren Geschichte in Dialekten, Gesängen, Wanderrouten und mütterlichen Linien—also in verkörperten, akustischen Trägern, die mit ihren Trägern sterben und sich daher nicht zu dauerhafter, ablösbarer Technik verdichten. Das sogenannte "Archäozeten–Paradox" ist zu großen Teilen ein Beobachtungsfehler: Wer nur Stein, Feuer und Knochen zu lesen versteht, überhört die Zeit, die in Klang, Strömung, Verwandtschaft und Wanderung aufbewahrt ist. Die Gletscherzyklen verschärfen diese Asymmetrie, denn die großen marinen Linien werden an den Ressourcentälern kulturell zurückgesetzt, während terrestrische Linien ihren Artefaktspeicher durch die Kältephasen retten.

9 Empirical anchors

The framework maps onto: gigantic cetaceans (*Basilosaurus*, *Livyatan melvillei*, *Physeter macrocephalus*); giant dolphins (*Ankylorhiza*, *Pebanista*, and the extinct platanistoid radiations); and shark controls (*Otodus megalodon*, the *Carcharodon* lineage). Sperm–whale vocal clans span thousands of kilometres, are distinguished by dialect, and are maintained by matrilineal social learning [3–5]; humpback song propagates and culturally evolves across ocean basins [7, 8]; resident killer–whale call types change over decades within stable matrilines [10, 11]. These are the direct empirical signatures of strong H_c, H_e with weak H_a.

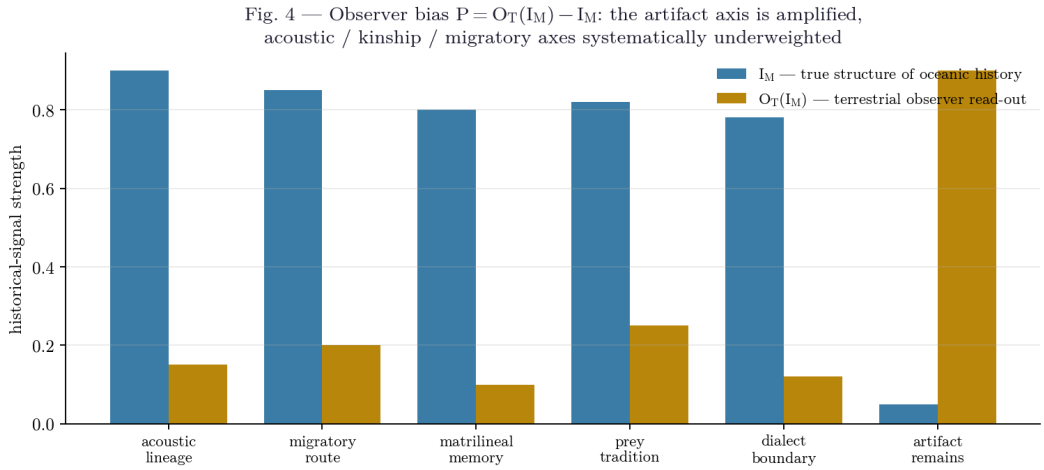


Figure 4: Observer bias $P = O_T(I_M) - I_M$. Acoustic lineage, migratory route, matrilineal memory, prey tradition and dialect boundaries carry strong real historical signal, yet an artifact–centred operator systematically underweights them.

10 Falsifiable predictions

1. Higher marine intelligence should correlate with reliance on short–period social structure (matriline/clan) rather than long–term artifact accumulation.

2. Increasing gigantism ⇒ amplified energetic variance ⇒ reduced cultural stability, sharpened at glacial minima.

3. All marine intelligence systems converge on transient embodied–acoustic culture rather than cumulative civilisation.

4. If any lineage is observed to place and re–use durable external substrate across generations, its C_a should exceed the bounded ceiling predicted here—a direct route to falsification.

11 Conclusion: from "suppressed history" to "divergent media"

The old proposition—"the ocean permits intelligence but systematically suppresses history"—should be discarded. The accurate proposition is:

> *Marine ecosystems preserve history but tend to compress it into body, sound, kinship networks, migratory routes and ecological–niche traditions; terrestrial environments more readily externalise history into tools, fire, material structures and symbolic systems. The core of the archaeocete paradox is not that cetaceans lack history, but that the principal carriers of cetacean history do not deposit in the forms most familiar to human archaeology.*

Formally, the central sentence of the framework is:

> *The cetacean paradox is not the absence of history in marine intelligence, but the divergence between embodied–acoustic historical continuity and artifact–based cumulative technological inheritance.*

Or, to close: the whales did not fail at history; what failed is the terrestrial observer who can read only stone, fire and bone, and does not know how to read the time preserved in sound, current, kinship and migration.

Note. All figures are generated by the accompanying unified numerical kernel (`models_v2.py`); the three models share one constraint parameter set, so that the ODE, ABM and phase–transition descriptions remain internally consistent. An interactive numerical companion is provided as HTML.

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